Hyperspectral Target Detection With RoI Feature Transformation and Multiscale Spectral Attention

Yanzi Shi, *Graduate Student Member, IEEE*, Jiaojiao Li, *Member, IEEE*, Yuxuan Zheng, *Graduate Student Member, IEEE*, Bobo Xi, *Graduate Student Member, IEEE*, and Yunsong Li, *Member, IEEE*

***Abstract*— Target detection plays a core issue in hyperspectral remote sensing, but faces serious challenges of how to deal with the spatial and spectral redundancies and spectral variations. In this article, a novel network block is developed, called RFT-MSA block (abbreviated as RM), which includes the region-of-interest (RoI) feature transformation (RFT) and the multiscale-spectral-attention (MSA) module as to reduce the spatial and spectral redundancies simultaneously and provide strong discrimination. Furthermore, a deep spatial–spectral network (DSSN) is presented by stacking several RM and deconvolutional (DC) blocks for hyperspectral target detection in an unsupervised manner, and a feature loss term is investigated to simultaneously restrict the target to be sparse and minimize the energy of the background. The proposed algorithm mainly consists of three steps. First, an RoI map is detected using a classical detector (no statistic assumption is needed) with an edge-preserving filter. Then, the hyperspectral image (HSI) and the corresponding RoI map are considered as inputs to the DSSN for extracting the spatial and spectral feature of interest (SSFI). Finally, we apply the nearest neighbors (NNs) to the SSFI for detection-map refinement. The experimental results on one synthetic and three real HSIs demonstrate that the proposed algorithm outperforms other benchmark approaches in detection performance and robustness. In addition, further analysis also demonstrates the effectiveness of the proposed RM block.**

***Index Terms*— Feature loss, hyperspectral target detection, multiscale spectral attention (MSA), region of interest (RoI) feature transformation (RFT), spatial and spectral feature of interest (SSFI).**

Manuscript received March 25, 2020; revised May 17, 2020; accepted June 8, 2020. Date of publication July 29, 2020; date of current version May 21, 2021. This work was supported in part by the National Key Research and Development Program of China under Grant 2018AAA0102702, in part by the National Nature Science Foundation of China under Grant 61901343, in part by the China Postdoctoral Science Foundation under Grant 2017M623124, in part by the China Postdoctoral Science Special Foundation under Grant 2018T111019, in part by the Open Research Fund of the CAS Key Laboratory of Spectral Imaging Technology under Grant LSIT201924W, in part by the Fundamental Research Funds for the Central Universities under Grant JB190107, in part by the National Nature Science Foundation of China under Grant 61571345, Grant 61671383, Grant 91538101, Grant 61501346, and Grant 61502367, in part by the 111 Project under Grant B08038, and in part by the Innovation Fund of Xidian University under Grant 5001-20109195456. *(Corresponding authors: Jiaojiao Li; Yunsong Li.)*

Yanzi Shi, Yuxuan Zheng, Bobo Xi, and Yunsong Li are with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China (e-mail: yzshi_xidian@163.com; yxzheng24@163.com; xibobo1301@foxmail.com; ysli@mail.xidian.edu.cn).

Jiaojiao Li is with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China, and also with the CAS Key Laboratory of Spectral Imaging Technology, Xi'an 710119, China (e-mail: jjli@xidian.edu.cn).

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Digital Object Identifier 10.1109/TGRS.2020.3001948

I. Introduction

HYPERSPECTRAL images (HSIs) can capture a wide range of wavelengths with high spectral resolution, which is up to nanometer level. The rich spectral information makes it possible to distinguish subtle spectral discrepancies from the spatially similar objects, which can hardly be differentiated by the natural and multispectral images. Target detection as a key task has drawn increasing attention in HSI processing. There are many applications that require to detect and recognize the targets in a specific scene, such as mineral exploration, vegetation monitoring, toxin tracking, methane detection, and nuclear threat identification [1]–[4].

Over the past decades, many target-detection algorithms have been developed. Among them, the well-known statistics-based methods are most frequently employed, including the adaptive coherence estimation (ACE) [5], adaptive matched filter (AMF) [6], matched subspace detector (MSD) [7], adaptive subspace detector (ASD) [8], and hybrid structured detector (HSD) [9]. In addition, the orthogonal subspace projection (OSP) [10] constructs a vector to maximize the signal-to-noise ratio (SNR) to annihilate the interference component obtained by endmember extraction or prior knowledge. However, the aforementioned classical models only work well in specific situations, since they are all based on the assumption that the background/interference confirms to a multivariate Gaussian distribution, which is generally not valid in reality. There are some representative methods that make no assumption on the background/interference statistics, e.g., constrained energy minimization (CEM) [11] and sparse representation (ST)-based methods. CEM designs a finite-impulse-response (FIR) linear filter, maximizing the filter's response to the target while suppressing that to the background. The ST-based detectors represent a given test pixel as a linear combination of a very few target and background atoms obtained from an overcomplete dictionary, and the residual between the reconstruction error of the target and background atoms can represent the class information of the test pixel. Famous algorithms include the sparsity-based target detector (STD) [12], sparse representation-based binary hypothesis detector (SRBBHD) [13], and joint sparse representation and multitask learning (JSR-MTL) [14]. The major disadvantage of the detectors mentioned above is that they are mainly based on the low-order statistics and cannot describe the nonlinearity of the HSI better. To address this problem, kernel tricks are used to reveal the nonlinear characteristics of the HSI [15]–[19].

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In recent years, data-driven deep learning has shown promising performance in HSI processing, such as pansharpening [20], [21], denoising [22], [23], anomaly detection [24], [25], unmixing [26], [27], and classification [28]–[30]. For hyperspectral target detection, deep learning gradually emerges as a powerful tool. However, the main problem of deep learning in target detection is that only a few target samples are available. One typical solution is to enlarge the training samples. In [31], a multilayer convolutional neural network (CNN)-based target-detection framework was proposed, which can enlarge the sample size through generating the differences between the pixel pair features (PPFs) of the reference data set. In [32], a subtraction PPF method was proposed to further enlarge the training samples, which considers a target class to ensure that the CNN is a detection function. Another solution is focusing on unsupervised learning, which allows mining high-level features for target detection with unlabeled data. Our recent works that employed autoencoder (AE) for discriminative feature extraction and target detection also achieved promising performance [33], [34]. However, we note that the existing deep-learning-based works can be hampered by equalizing the contribution of each band and pixel, since some pixels and bands are useless and redundant. Specifically, the spectral redundancy may increase the intraclass dissimilarities and the interclass similarities, and the spatial redundancy makes the neural network pay much attention on the unimportant regions of noninterest. This fact imposes significant challenges on designing the effective target-detection methods.

To address the aforementioned challenges, this article proposes a novel deep spatial–spectral network (DSSN) framework for hyperspectral target detection in an unsupervised manner. The proposed architecture composes of an RMs-subnet and a DCs-subnet stacked by several RFT-MSA (abbreviated as RM) and deconvolutional (DC) blocks, respectively. In detail, the RM block consists of two effective modules: region-of-interest (RoI) feature transformation (RFT) for spatial redundancy reduction and multiscale spectral attention (MSA) for spectral redundancy removal, and the DC block includes two deconvolutional layers for HSI reconstruction. Here are two main modules of this article. First, the RFT module takes an RoI map as the input to generate a pair of modulation parameters, based on which it applies affine transformation spatially to the HSI/feature map. In this way, the network is enforced to focus on the RoIs mainly including the target and the hard-detected background pixels. Second, the MSA module is adopted to distinct further the target and hard-detected background samples, which leverages the merits from two aspects.

1) *Multiscale Spectral–Spatial Feature Extraction (MS^{2}FE):* Taking the output of the RFT module as the input, MS^{2}FE first learns the multiscale spectral–spatial information with a hierarchical residual-like architecture [35], which further improves the discriminative capacity of the hidden features and, thus, boosts the accuracy of hyperspectral target detection.
2) *Band Attention (BA):* We design our own BA unit on the multiscale spectral–spatial features to adaptively remove useless bands that carry no discriminative information and, thus, yield gains on detection performance and computational cost.

Apart from the above two novel modules, two types of skip connections are introduced to ease the training of the DSSN, and a feature loss is investigated to further increase interclass differences. Hence, the whole network can extract the most discriminative spectral–spatial information, i.e., spatial and spectral feature of interest (SSFI), specifically for hyperspectral target detection. In a nutshell, the main contributions of this article can be summarized as follows.

1) The DSSN architecture based on unsupervised learning is specifically designed to extract discriminative spectral–spatial features for hyperspectral target detection, which requires only one target sample and no human interactions.
2) The RM block is proposed to transform the original HSI into SSFI, which contributes differently to target detection in different pixels and bands. More importantly, the RM block can be integrated into other standard architectures for HSI processing, such as classification and anomaly detection.
3) From the viewpoint of network training, the shortcut and long-term connections are applied to avoid the notorious gradient vanishing/exploding problem and ease network training, respectively. As for the detection task, the feature loss consisting of ℓ_1-norm and ℓ_2-norm regularizations is presented to obtain more discriminative feature.

It is worth noting that the proposed DSSN is more suitable for pixel-level HSI target detection. For subpixel-level target detection with little spatial information, the DSSN may not perform as well as that for pixel-level target detection, but can also obtain satisfying performance. The difference between the pixel-level and subpixel-level target detections in this work is in the MS^{2}FE unit. Since the subpixel target contains few shape and textural information, the discriminative feature learned by the MS^{2}FE unit is dominant by spectral information rather than by spatial information. In this case, the MS^{2}FE unit is substantially a kind of multiscale spectral feature extractor.

The remainder of this article is organized as follows. Section II briefly reviews the CEM detector and attention mechanism, and gives the main notations adopted throughout this article. In Section III, we introduce the proposed DSSN model in detail. Section IV shows the experimental studies and further analysis for verifying the DSSN. Finally, Section V draws the conclusion of this article.

II. Related Work

A. Summary of Notations

A summary of the main notations adopted in this article is listed in Table I.

B. CEM

Given an HSI noted $\mathbf{X}$ with $N_h \times N_w$ pixels and N_λ bands, and a prior-known target spectrum $\mathbf{x}_t \in \mathbb{R}^{N_\lambda \times 1}$, the objective of CEM is to build an FIR filter $\mathbf{w} \in \mathbb{R}^{N_\lambda \times 1}$ that minimizes

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TABLE I
SUMMARY OF THE MAIN NOTATIONS ADOPTED THROUGHOUT THIS ARTICLE

Notations	Meanings
$\mathbf{X} \in \mathbb{R}^{N_h \times N_w \times N_\lambda}$	Hyperspectral data cube
$\mathbf{R} \in \mathbb{R}^{N_h \times N_w}$	RoI map
N_p	Spatial size of training patch
$\mathbf{X}_t \in \mathbb{R}^{N_p \times N_p \times N_\lambda}$	Target patch of HSI
$\mathbf{R}_t \in \mathbb{R}^{N_p \times N_p}$	Target patch of RoI map
K	Number of neurons in fully connection layer
$\mathbf{Y} \in \mathbb{R}^{N_h \times N_w \times K}$	Spatial and spectral feature of interest (SSFI)
$\hat{\mathbf{X}} \in \mathbb{R}^{N_h \times N_w \times N_\lambda}$	Reconstructed HSI cube
$\mathbf{\Theta}_{3d}$	Weights of the RMs-subnet
$\hat{\mathbf{\Theta}}_{3d}$	Weights of the DCs-subnet
$\zeta(\mathbf{X}, \mathbf{R}, \mathbf{\Theta}_{3d})$	Nonlinear mapping function of the RMs-subnet
$\hat{\zeta}(\mathbf{Y}, \hat{\mathbf{\Theta}}_{3d})$	Nonlinear mapping function of the DCs-subnet

the output energy under the constraint $\mathbf{w}^T\mathbf{x}_t = 1$. Specifically, the problem can be formulated as

$$\min_{\mathbf{w}} \frac{1}{N_{\text{hw}}} \sum_{i=1}^{N_{\text{hw}}} \left(\mathbf{w}^T\mathbf{x}_i\right)^2 = \min_{\mathbf{w}} \mathbf{w}^T \Phi \mathbf{w} \quad \text{s.t. } \mathbf{w}^T\mathbf{x}_t = 1 \tag{1}$$

where $N_{\text{hw}} = N_h \times N_w$, $\mathbf{x}_i$ is the ith pixel vector of the image and $\Phi = (1/N_{\text{hw}}) \sum_{i=1}^{N_{\text{hw}}} \mathbf{x}_i\mathbf{x}_i^T$ denotes the autocorrelation matrix of the image. The optimal solution of (1) can be derived in [36] as follows:

$$\mathbf{w}_{\text{CEM}} = \frac{\Phi^{-1}\mathbf{x}_t}{\mathbf{x}_t^T\Phi^{-1}\mathbf{x}_t}. \tag{2}$$

With the optimal filtering vector obtained by (2), the detection result of $\mathbf{w}_{\text{CEM}}$ from the ith pixel is

$$c_i = \mathbf{w}_{\text{CEM}}^T\mathbf{x}_i, \quad i = 1, 2, \ldots, N_{\text{hw}}. \tag{3}$$

The detection map of CEM is referred to as $\mathbf{C} = [c_1\ c_2\ \cdots\ c_{N_{\text{hw}}}]$.

C. Attention Mechanism

In recent years, the attention mechanism has been demonstrated effectively and widely used in the community of natural language processing (NLP) [37], visual question answering (VQA) [38], and computer vision (CV) [39], and it allows neural networks to allocate more attention to the more informative regions. According to [40], the attention mechanism can be divided into two categories: soft and hard. The soft attention is a fully differentiable deterministic mechanism that can be plugged into an existing system, while the hard attention is a stochastic and nondifferentiable process, and the gradients are estimated by Monte Carlo sampling. In this article, we steer our focus on the soft model, since the gradient can be directly computed through backpropagation (BP). There are three types of soft-attention modules: spatial attention [41], channel attention [39], and mixed attention [42]. As for the spectral redundancy reduction in the HSI, channel attention is particularly appropriate to recalibrate the weight of different spectral channels. Inspired by the SE block [39], we devise our BA unit for hyperspectral target detection, which is lightweight and computationally efficient.

III. METHODOLOGY

A. Proposed Framework Description

This article presents a novel DSSN for hyperspectral target detection with the RFT and MSA strategies to overcome the shortcomings of the existing approaches. Fig. 1 demonstrates the flowchart of our proposed framework. The main objectives of the DSSN are to reject the easily detected background pixels of noninterest in the RoI map and focus on the target pixels of interest, i.e., SSFI extraction. The proposed DSSN consists of three RM and DC blocks, and two fully connection layers. The inputs of the DSSN are the HSI and the corresponding RoI map, while the expected output is the input HSI itself. Here, we pay attention to the SSFI learned by the RMs-subnet rather than the reconstructed output that does not contain any additionally discriminative information compared with the original HSI. Moreover, considering that the extracted SSFI focuses more on the target characteristics and may lose the majority information of the background, the long-term connections between the mirrored blocks are adopted to ease the reconstruction of the lost background.

With the notations demonstrated in Table I, the DSSN takes the HSI patch $\mathbf{X}_i \in \mathbb{R}^{N_p \times N_p \times N_\lambda}$ and the RoI patch $\mathbf{R}_i \in \mathbb{R}^{N_p \times N_p}$ as the inputs. The extracted high-level feature vector $\mathbf{y}_i \in \mathbb{R}^{K \times 1}$ and the reconstructed patch $\hat{\mathbf{X}}_i$ are calculated as follows:

$$\mathbf{y}_i = \zeta(\mathbf{X}_i, \mathbf{R}_i, \mathbf{\Theta}_{3d}) \tag{4}$$

$$\hat{\mathbf{X}}_i = \hat{\zeta}\left(\mathbf{y}_i, \hat{\mathbf{\Theta}}_{3d}\right). \tag{5}$$

The SSFI is represented as $\mathbf{Y} = [\mathbf{y}_1\ \mathbf{y}_2\ \cdots\ \mathbf{y}_{N_{\text{hw}}}]$. Similarly, the target vector can be written as

$$\mathbf{y}_t = \zeta(\mathbf{X}_t, \mathbf{R}_t, \mathbf{\Theta}_{3d}). \tag{6}$$

Following [43], the Student's t-distribution with a degree of freedom υ (set to 1 in practice) is adopted to measure the distance between the target vector and the SSFI in the embedding feature space, which is defined as follows:

$$\gamma_i = \left(1 + \frac{\|\mathbf{y}_i - \mathbf{y}_t\|_2^2}{\upsilon}\right)^{-\frac{\upsilon+1}{2}}. \tag{7}$$

The loss function of the DSSN is designed through the following calculation:

$$L = \frac{1}{N_b} \sum_{i=1}^{N_b} \|\mathbf{X}_i - \hat{\mathbf{X}}_i\|_F^2 + \lambda_1(\|\mathbf{\Gamma}\|_1 + \|\mathbf{\Gamma}\|_2) + \frac{\lambda_2}{2N_b}\|\mathbf{\Theta}\|_2^2 \tag{8}$$

where N_b is the minibatch size, $\mathbf{\Theta} = [\mathbf{\Theta}_{3d}\ \hat{\mathbf{\Theta}}_{3d}]$, $\mathbf{\Gamma} = [\gamma_1\ \gamma_2\ \cdots\ \gamma_{N_b}]$, $\|\mathbf{\Gamma}\|_1$ is used for the sparsity formulation of the target, and $\|\mathbf{\Gamma}\|_2$ for the energy minimization of the

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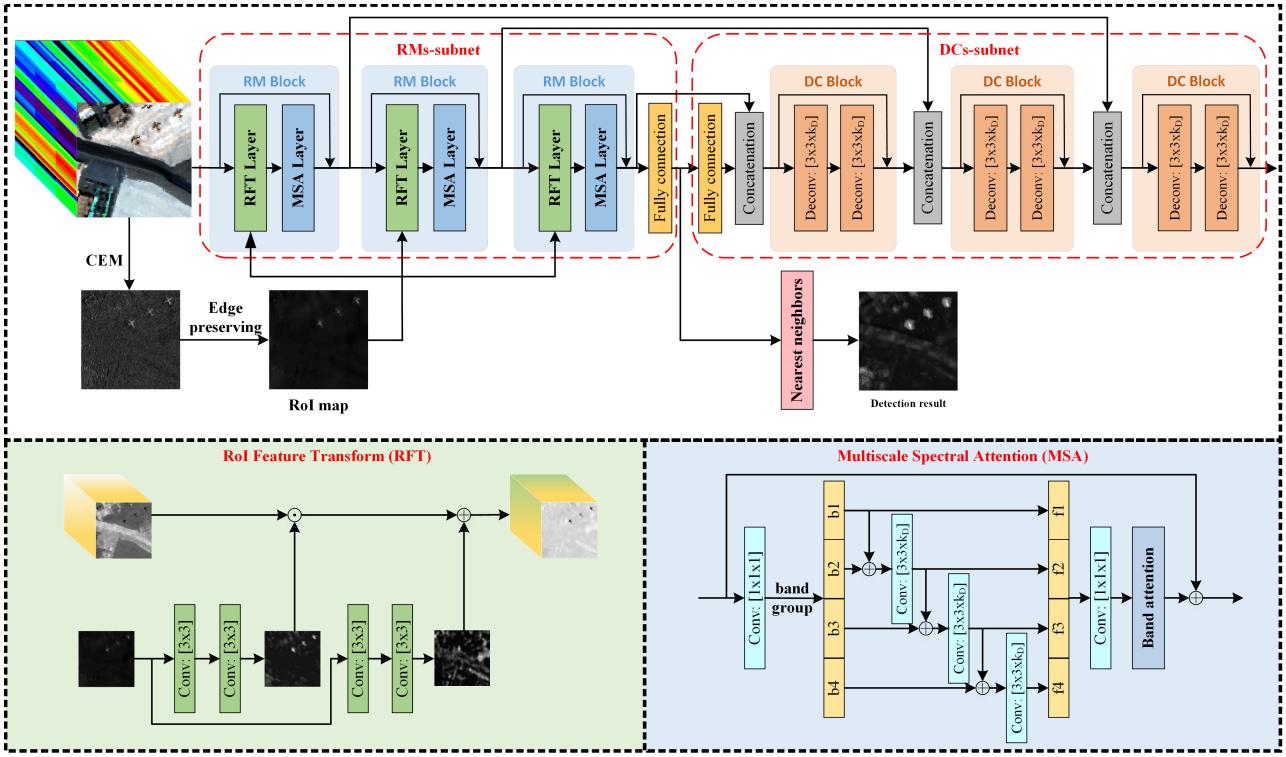

Fig. 1. Flowchart of the proposed method.

background. Parameters λ_1 and λ_2 control the importance of the feature loss term and the regularization term, respectively. Minimizing formulation (8) fosters the DSSN to extract features that can produce the sparse detection map with relatively low energy, i.e., preserves target samples while suppressing the background samples.

At last, the final detection map is obtained by the nearest neighbors (NNs), namely, calculating the Euclidean distance between SSFI $\mathbf{Y}$ and the target vector $\mathbf{y}_t$ as the detection value.

B. Construction of RoI Map

The construction of the RoI map is formed by two steps: 1) spectral detection by CEM and 2) smoothing by an edge-preserving filter. From a practical point of view, CEM is adopted for spectral detection, because it only requires the target spectrum without making any assumptions of data distribution. The spectral detection map $\mathbf{C}$ is determined by (3).

In the HSI target detection, the spectral mixture is problematic, especially the target pixels at the edge area may be omitted with high probability. To address this problem, we adopt edge-preserving filtering to smooth $\mathbf{C}$ and decrease the differences between the subpixels at the edge and the central pure pixels. The domain transform recursive filter (DTRF) due to its real-time process proposed in [44] is selected for edge-preserving, in which a 2-D edge-preserving filter can be implemented by a sequence of vertical and horizontal 1-D filters. Take horizontal filtering as an example; given a horizontal signal $\mathbf{C}_h$, the DTRF first computes the domain transform

$$d_n = \mathbf{C}_h^0 + \sum_{j=1}^{n}\left(1 + \frac{\sigma_s}{\sigma_r}\left|\mathbf{C}_h^j - \mathbf{C}_h^{j-1}\right|\right) \tag{9}$$

where σ_s^2 and σ_r^2 are the variances of the DTRF over the spatial coordinates and the range coordinates (intensity), respectively. d_n is the signal at coordinate n in the transformed domain. Then, the smoothing output can be defined by a recursive filter in the transformed domain as

$$\mathbf{R}_h^n = \left(1 - a^b\right)\mathbf{C}_h^n + a^b\mathbf{R}_h^{n-1} \tag{10}$$

and

$$a = e^{-\sqrt{2}/\sigma_{H_k}} \tag{11}$$

$$\sigma_{H_k} = \sigma_H\sqrt{3}\frac{2^{N_l-k}}{\sqrt{4^{N_l}-1}} \tag{12}$$

$$\sigma_H = \sigma_s \tag{13}$$

$$b = d_n - d_{n-1} \tag{14}$$

in which N_l is the number of iterations, σ_{H_k} is the standard deviation of the kth iteration, and $\mathbf{R}_h^n$ is the filtering $\mathbf{C}_h$ at coordinate n. The final RoI map is obtained by interleaving vertical and horizontal filterings in several iterations. For the mixed pixels in the edge, a^b is in $(0, 1)$, fostering the propagation of the neighboring pure pixels and, thus, improving the detection probability without regarding the false-alarm rate. The source code of the DTRF is available on Professor Gastal's homepage.[1]

[1]http://www.inf.ufrgs.br/~eslgastal/

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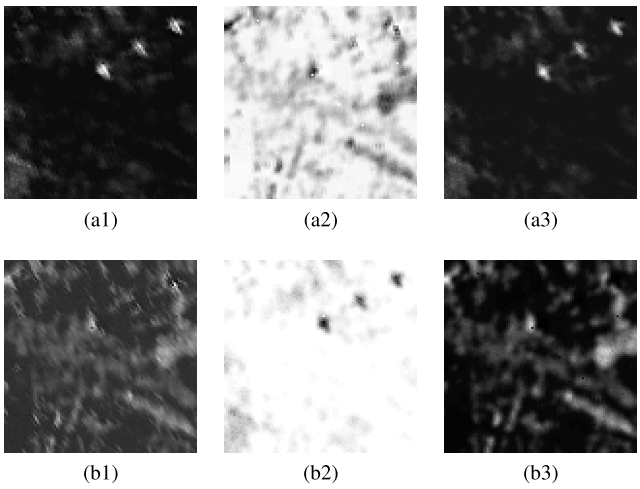

Fig. 2. Modulation parameter visualization from three RM blocks after training on the San Diego data set. (a1) $\mathbf{M}_1$. (a2) $\mathbf{M}_2$. (a3) $\mathbf{M}_3$. (b1) $\mathbf{A}_1$. (b2) $\mathbf{A}_2$. (b3) $\mathbf{A}_3$.

C. RM Block

The RM block consists of an RFT layer and an MSA layer, which can be employed directly to other deep neural networks as an independent block. It enforces the network to concentrate on the target regions from the spatial–spectral point of view. The detailed descriptions of the RFT layer and the MSA layer are demonstrated as the following contents.

1) RFT: The RFT layer focuses on the undesired background elimination, which automatically learns a transform function χ to generate a pair of modulation parameters through a simple neural network with four convolution layers. Specifically

$$\left(\mathbf{M}_j, \mathbf{A}_j\right) = \chi_{\mathrm{j}}(\mathbf{R}) \tag{15}$$

in which $\mathbf{M}_j$ and $\mathbf{A}_j$ are the transformation parameters in the jth RM block, $j = 1, 2, 3$. Then, the feature transformation is carried out as follows:

$$\mathbf{H}_j = \mathbf{M}_j \cdot \mathbf{I}_j + \mathbf{A}_j \tag{16}$$

where $\mathbf{I}_j$ and $\mathbf{H}_j$ denote the input and output of the RFT layer in the jth RM block, respectively. Operations $\cdot$ and $+$ both support broadcasting across spectral dimension. For clarity, only one channel of the feature map is depicted at the left bottom of Fig. 1.

In order to understand better the effectiveness of the RFT layer, Fig. 2 visualizes the modulation parameters in three RM blocks. The visualization reveals that the two parameters demonstrate extreme difference between the interested airplane and the undesired background in a complementary way. The feature maps of $\mathbf{A}_1, \mathbf{M}_2, \mathbf{A}_3$ are less distinguished than that of $\mathbf{M}_1, \mathbf{A}_2, \mathbf{M}_3$, since the objective of the DSSN is not only the discriminative feature extraction but also the preservation of the intrinsic data property.

2) MSA: Multiscale feature representation recently draws increasing attention in HSI classification. For example, Li *et al.* [45] proposed a different kernel size strategy to capture the multiscale spatial features. Zhao and Du [46] automatically learned multiscale representations by constructing multiscale training samples with the image pyramid algorithm. Different from the aforementioned methods, we learn the multiscale spectral–spatial feature only through a novel MSA block in a more granular way.

Fig. 1 (right bottom) gives the architecture of the MSA module. It can be seen that the basic structure is the residual block. Different from the standard residual learning, the proposed residual branch consists of several small filter groups that are connected in a hierarchical style. Specifically, the input feature of the jth RM block $\mathbf{H}_j$ is first divided into g subgroups with equal size by the following band-grouping method:

$$\begin{aligned}
\mathbf{b}_j^1 &= \left[\mathbf{H}_j^1, \mathbf{H}_j^{1+g}, \cdots, \mathbf{H}_j^{1+(m-1)g}\right] \\
\mathbf{b}_j^2 &= \left[\mathbf{H}_j^2, \mathbf{H}_j^{2+g}, \cdots, \mathbf{H}_j^{2+(m-1)g}\right] \\
&\vdots \\
\mathbf{b}_j^g &= \left[\mathbf{H}_j^g, \mathbf{H}_j^{2g}, \cdots, \mathbf{H}_j^{mg}\right]
\end{aligned} \tag{17}$$

where $m = \lfloor G/g \rfloor$ and G is the total number of $\mathbf{H}_j$ in band dimension. Except that the first feature subgroup $\mathbf{b}_j^1$ is directly connected to the output, each subgroup is followed by a convolution with a filter of $3 \times 3 \times k_D$ and a stride of 1, where k_D is the kernel size along the spectral dimension. For the feature subgroup $\mathbf{b}_j^i, i \geq 2$, it is first added to the output feature map of the previous subgroup $\mathbf{b}_j^{i-1}$ and then convolved with the $3 \times 3 \times k_D$ kernel. Specifically, the output of each subgroup performs the following calculation:

$$\mathbf{f}_j^i = \begin{cases} \left(\mathbf{f}_j^{i-1} + \mathbf{b}_j^i\right) \otimes \boldsymbol{\omega}_j^{3\times 3\times k_D}, & 2 \leq i \leq g \\ \mathbf{b}_j^i, & i = 1 \end{cases} \tag{18}$$

where $\otimes$ denotes the convolution, and $\mathbf{f}_j^i$ and $\boldsymbol{\omega}_j^{3\times 3\times k_D}$ are the output and kernel of the ith subgroup. Finally, the outputs of all subgroups are concatenated and convolved with $\boldsymbol{\omega}_j^{1\times 1\times 1}$ of stride $1 \times 1 \times 2$ as follows:

$$\mathbf{F}_j = \left[\mathbf{f}_j^1 \quad \mathbf{f}_j^2 \quad \cdots \quad \mathbf{f}_j^g\right] \otimes \boldsymbol{\omega}_j^{1\times 1\times 1} \tag{19}$$

in which $\mathbf{F}_j$ denotes the extracted multiscale feature of the jth RM block. According to (18), we can see that the receptive field of the previous subgroup increases whenever it goes through a convolution filter, generating multiscale spectral–spatial features.

Moreover, the attention mechanism is integrated with the MS^{2}FE for adaptively selecting the most informative bands. Expanding on the prior SENet [39], we devise our own BA module that is more appropriate for HSI processing. Formally, a weight vector $\boldsymbol{\mu}_j$ is initialized by global average pooling through the spatial and channel dimensions, shown as follows:

$$\boldsymbol{\mu}_j = \frac{1}{C \times H \times W} \sum_{c=1}^{C} \sum_{h=1}^{H} \sum_{w=1}^{W} \mathbf{F}_j^{\mathrm{chw}} \tag{20}$$

where $\boldsymbol{\mu}_j \in \mathbb{R}^{D_j \times 1}$, and C, H, W, D_j denote the channel, height, width, and band of $\mathbf{F}_j$, respectively. Then, we parameterize $\boldsymbol{\mu}_j$ with two dense layers to further capture the bandwise dependences

$$\boldsymbol{v}_j = \delta_{\mathrm{sigmoid}}\left(\boldsymbol{\omega}_2 \otimes \delta_{\mathrm{relu}}\left(\boldsymbol{\omega}_1 \otimes \boldsymbol{\mu}_j\right)\right) \tag{21}$$

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in which $\boldsymbol{v}_j$ denotes the optimized weight vector, $\boldsymbol{\omega}_1 \in \mathbb{R}^{(D_j/r)\times D_j}$ and $\boldsymbol{\omega}_2 \in \mathbb{R}^{D_j\times(D_j/r)}$ are the weight matrices, and r is the reduction ratio. Further analysis of $\boldsymbol{v}_j$ is given in Section IV. The final output of the MSA module is calculated by broadcasting multiplication across spectral dimension as follows:

$$\mathbf{O}_j = \boldsymbol{v}_j \cdot \mathbf{F}_j. \tag{22}$$

The RMs-subnet is built with three RM blocks and one fully connection layer, in which each RM employs shortcut connection to avoid the notorious gradient vanishing/exploding problem. Thus, the input of the $(j+1)$th RM block is calculated as follows:

$$\mathbf{I}_{j+1} = \mathbf{O}_j + \mathbf{I}_j. \tag{23}$$

D. DC Block

The DC block is composed of two deconvolution layers, the kernel sizes of which are $3\times3\times k_D$ with stride $1\times1\times1$ and $3\times3\times k_D$ with stride $1\times1\times2$, respectively. Like the RM block, shortcut connection is also adopted to the DC block. To reconstruct the input HSI from the background lost SSFI better, a concatenation of hidden features from the RM and symmetric DC blocks is employed.

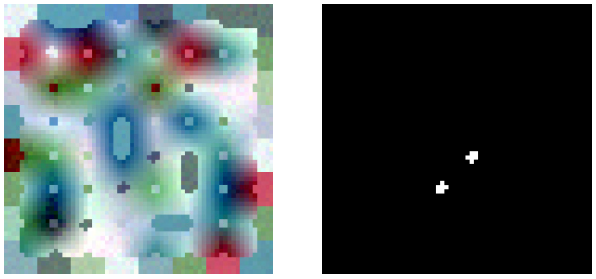
IV. Experiments and Analysis

A. Data Description

1) Synthetic Data: The first synthetic data is provided in [47] and have been used widely for target detection in [47] and [48]. It is generated by implanting 15 target spectra obtained from the United States Geological Survey (USGS) digital spectral library [49]. Among the 15 spectra, Labradorite HS17.3B is selected as the desired target. This data consist of 64×64 pixels with 224 bands. Moreover, a Gaussian noise with 20 dB is added to the synthetic image for reality. Fig. 3 displays the pseudocolor image of the data and its ground truth.

2) San Diego Data: The second data was collected using the Airborne Visible/InfraRed Imaging Spectrometer (AVIRIS) over the San Diego airport, CA, USA. It contains 100×100 pixels with a spatial resolution of 3.5 m and 189 bands ranging from 400 to 2500 nm with a spectral resolution of 10 nm after discarding 35 low SNR and water absorption bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224). The targets of interest are three airplanes. Fig. 4 shows the pseudocolor image and ground truth of the San Diego data.

3) Avon Data: The third data we employed was captured by a hyperspectral pushbroom ProSpecTIR-VS sensor, covering the 400–2450-nm wavelengths, over Avon, NY, USA, in the morning on September 20, 2012 [50]. A subimage of size $100\times100\times360$ with a spatial resolution of 1 m/pixel and a spectral resolution of 5 nm/band is cropped in this article, as shown in Fig. 5(a). There are two types of blue and brown tarps scattered in the scene. Among them, 12 blue tarps are selected as the targets to be detected, and the ground truth is shown in Fig. 5(b).

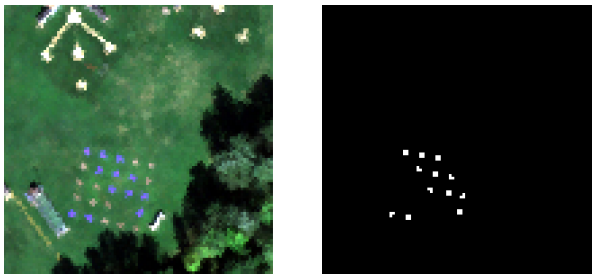


(a) (b)

Fig. 3. Synthetic data. (a) Pseudocolor image. (b) Ground truth.

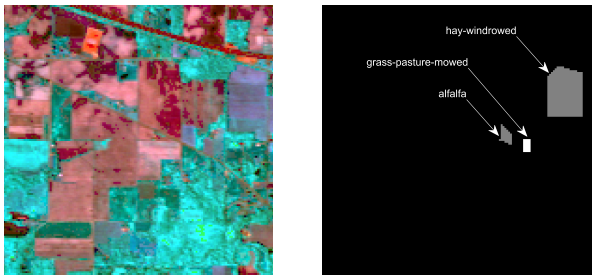
(a) (b)

Fig. 4. San Diego data. (a) Pseudocolor image. (b) Ground truth.



(a) (b)

Fig. 5. Avon data. (a) Pseudocolor image. (b) Ground truth.



(a) (b)

Fig. 6. Indian Pines data. (a) Pseudocolor image. (b) Ground truth.

4) Indian Pines Data: The fourth data was also acquired by the AVIRIS sensor over Indiana, USA. This image has 145×145 pixels with a spatial resolution of 20 m and 200 bands covering from 400 to 2500 nm after removing a low SNR and water absorption bands (104–108, 150–163, and 220–224). The grass-pasture-mowed class is selected as the target of

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interest, which is similar to the alfalfa and hay-windrowed classes measured by the spectral angle mapper (SAM). Fig. 6 shows the pseudocolor image and the corresponding ground truth.

B. Experimental Setup

1) Evaluation Criteria: In this article, we evaluate the performance of different detection methods from both subjective and objective perspectives. The subjective evaluation is given by the visual inspection of the detection maps. The 3-D receiver operating characteristic (3D-ROC) [51] and the corresponding 2D-ROC curves are adopted for quantitative evaluation. The 3D-ROC curve is generated with three axes specified by the threshold (τ), false-alarm rate ($\mathbf{P}_f$), and detection probability ($\mathbf{P}_d$), which can be calculated as follows:

$$\mathbf{P}_d(\tau) = \frac{\mathrm{TP}_\tau}{\mathrm{TP}_\tau + \mathrm{FN}_\tau}, \quad \mathbf{P}_f(\tau) = \frac{\mathrm{FP}_\tau}{\mathrm{FP}_\tau + \mathrm{TN}_\tau} \tag{24}$$

where TP_τ, FP_τ, FN_τ, and TN_τ denote the number of correct detection target pixels, false-alarm pixels, omissive detection target pixels, and correct detection background pixels under the threshold τ, respectively. In addition, the minimal $\mathbf{P}_f$ under $\mathbf{P}_d = 1$ (denoted as $\mathbf{P}_f^{\min}(\mathbf{P}_d = 1)$) is applied to verify the discrimination ability of different detectors. Moreover, the area under the 2D-ROC curve of $(\mathbf{P}_f, \mathbf{P}_d)$ (AUC) provides further evaluation of detection performance. The better the detection, the closer the 2D-ROC curve of $(\mathbf{P}_f, \mathbf{P}_d)$ is to the bottom-left corner, the closer the 2D-ROC curve of $(\tau, \mathbf{P}_d)$ is to the top-right corner, the closer the 2D-ROC curve of $(\tau, \mathbf{P}_f)$ is to the bottom-left corner, the smaller the $\mathbf{P}_f^{\min}(\mathbf{P}_d = 1)$, the larger the AUC.

2) Parameter Setting and Network Training: The parameter sensitivity in the proposed DSSN method is analyzed in this section. The main ten parameters, i.e., spatial size of training patch N_p, σ_r, and σ_s of DTRF, λ_1 and λ_2 of the loss function, learning rate, size of the extracted feature SSFI in spectral dimension K, reduction ratio r, minibatch size N_b, and kernel size along the spectral dimension k_D are evaluated in our network. The N_p and k_D are estimated from the size of the detected targets and set to $3, 7, 3, 7$ and $3, 3, 7, 3$ for the four data sets, respectively. For other parameters, we tested them in different ranges and find that the detection results are not sensitive to the magnitude of σ_r, σ_s, λ_1, λ_2, r, and N_b. Therefore, we fix these parameters to $0.2, 4, 0.1, 0.1, 4, 256$ in all experiments. The optimal learning rate and feature dimension K are found by the grid-search method, in the ranges $\{0.001, 0.005, 0.01, 0.05, 0.1\}$ and $\{10, 20, 30, 40, 50\}$, respectively. According to the detection results, the learning rates and K are set to 0.05, 0.01, 0.01, and 0.01 and 20, 20, 20, and 40 for the four tested data sets, respectively. Fig. 7 shows the effect of K on the detection performance of the optimal learning rate. It can be seen that SSFI with very much or few information cannot achieve the best outcome. Fewer features are not enough to represent the discrimination between the target and the background, while more features will lead to redundancy and, thus, weaken the detection performance.

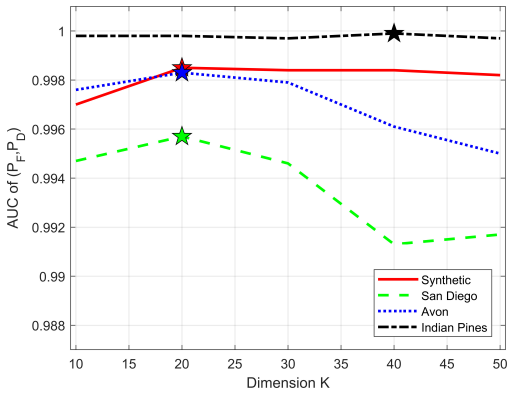

Fig. 7. Effect of parameter K on the detection performance.

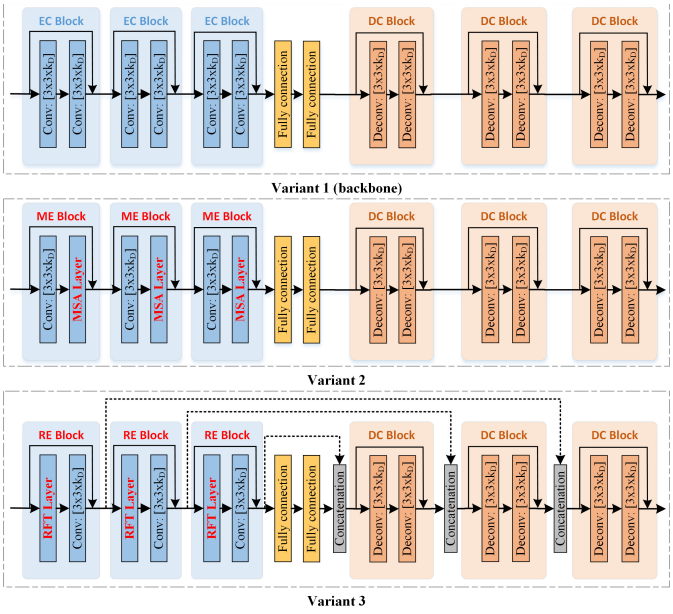

Fig. 8. Variants of DSSN for ablation analysis.

TABLE II
ABLATION ANALYSIS ON SAN DIEGO DATA. ARROW (↑) INDICATES THAT THE LARGER THE VALUE, THE BETTER THE RESULTS

Variant	RFT?	MSA?	AUC ↑
Variant 1 (backbone)	×	×	0.9325
Variant 2	×	✓	0.9696
Variant 3	✓	×	0.9920
Variant 4 (DSSN)	✓	✓	0.9957

C. Ablation Analysis

Ablation analysis is conducted in this section to elaborate the contributions of the proposed RFT and MSA modules. There are three variants of the proposed DSSN, as shown in Fig. 8. Variant 1 (backbone), variant 2, and variant 3 replace the RFT and MSA modules in the DSSN with two convolution layers (EC block), one convolution layer and MSA module

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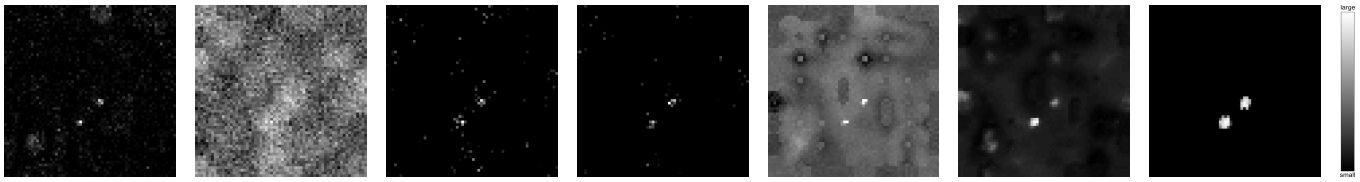
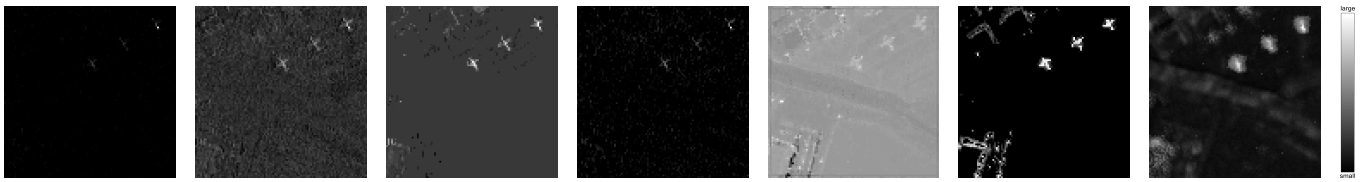


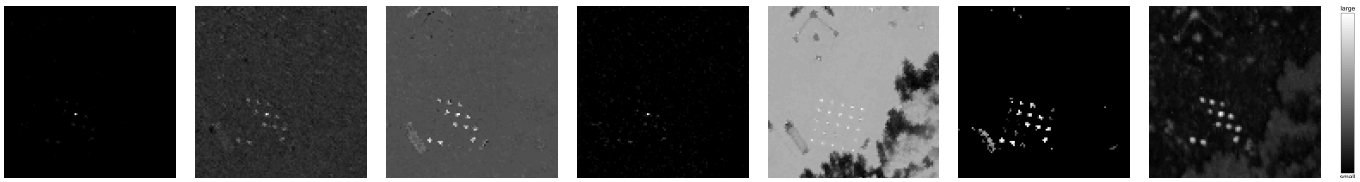
(a) (b) (c) (d) (e) (f) (g)

Fig. 9. Detection maps of different algorithms for the synthetic data. (a) ACE. (b) CEM. (c) hCEM. (d) E-CEM. (e) CSCR. (f) 3D-MMRAED. (g) DSSN.



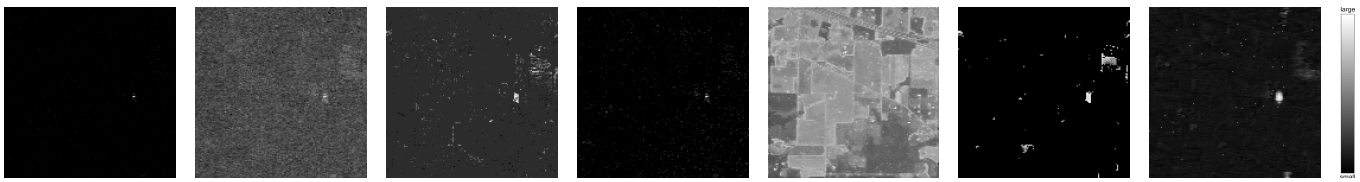
(a) (b) (c) (d) (e) (f) (g)

Fig. 10. Detection maps of different algorithms for the San Diego data. (a) ACE. (b) CEM. (c) hCEM. (d) E-CEM. (e) CSCR. (f) 3D-MMRAED. (g) DSSN.



(a) (b) (c) (d) (e) (f) (g)

Fig. 11. Detection maps of different algorithms for the Avon data. (a) ACE. (b) CEM. (c) hCEM. (d) E-CEM. (e) CSCR. (f) 3D-MMRAED. (g) DSSN.



(a) (b) (c) (d) (e) (f) (g)

Fig. 12. Detection maps of different algorithms for the Indian Pines data. (a) ACE. (b) CEM. (c) hCEM. (d) E-CEM. (e) CSCR. (f) 3D-MMRAED. (g) DSSN.

(ME block), and RFT module and one convolution layer (RE block), respectively. It should be noted that the long-term connection is only adopted when the RFT module is used. Table II gives the experimental results on the San Diego data. It can be seen that the performance of the backbone network without the RFT and MSA modules drops dramatically. Specifically, compared with the backbone, the detection accuracy obtained by variant 2 with MSA, variant 3 with RFT, and the DSSN with RFT and MSA increases to 3.71%, 5.95%, and 6.32%, respectively. The main reason is that RFT can reject most of the interference, and MSA selects the most informative bands, which helps to improve the discrimination capacity for target detection.

D. Results and Discussion

1) Compared With Different Target Detectors: Several detection algorithms are compared with our proposed DSSN for performance verification in the experiments, including four classical methods, which are ACE, CEM, the improved version of CEM, i.e., hierarchical CEM (hCEM)[2] [48] and ensemble based CEM (E-CEM)[2] [47], one representation-based method: combined sparse and collaborative representation (CSCR)[3] [52], and deep-learning-based 3D-MMRAED [34]. For the CEM and ACE methods, there are no parameters needed except for the prior known target signature, which remains the same in all competitors for each data set. For the hCEM detector, as in [48], two main parameters λ and ϵ are selected as 200 and 10^{-6} for all data sets, respectively. For E-CEM, four parameters, including the layer number, scanning windows, CEM number in each layer, and regularization parameter, are set as 10, $[(1/4)D, (2/4)D, (3/4)D, D]$, 6, and random$(0, 0.01)$ for all data sets, respectively, where D is the number of bands. For CSCR, the regularization parameters λ_1 and λ_2 are set as 10^{-1} and 10^{-2} for all data sets,

[2]http://levir.buaa.edu.cn/code_cn.htm
[3]http://cist.buct.edu.cn/staff/WeiLi/

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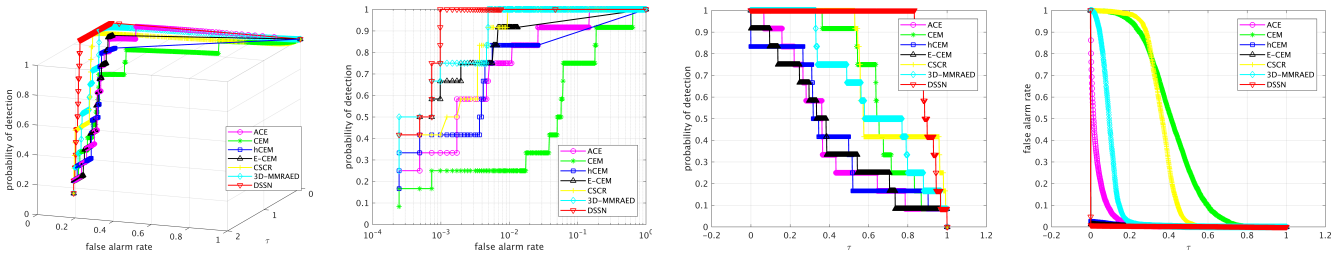

Fig. 13. 3D-ROC and the corresponding 2D-ROC curves of different algorithms for the synthetic data. (a) 3D-ROC. (b) 2D-ROC of $(\mathbf{P}_f, \mathbf{P}_d)$. (c) 2D-ROC of $(\tau, \mathbf{P}_d)$. (d) 2D-ROC of $(\tau, \mathbf{P}_f)$.

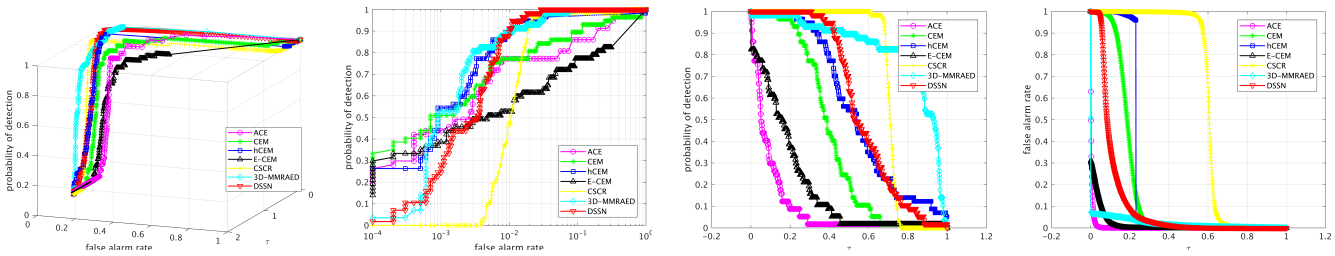

Fig. 14. 3D-ROC and the corresponding 2D-ROC curves of different algorithms for the San Diego data. (a) 3D-ROC. (b) 2D-ROC of $(\mathbf{P}_f, \mathbf{P}_d)$. (c) 2D-ROC of $(\tau, \mathbf{P}_d)$. (d) 2D-ROC of $(\tau, \mathbf{P}_f)$.

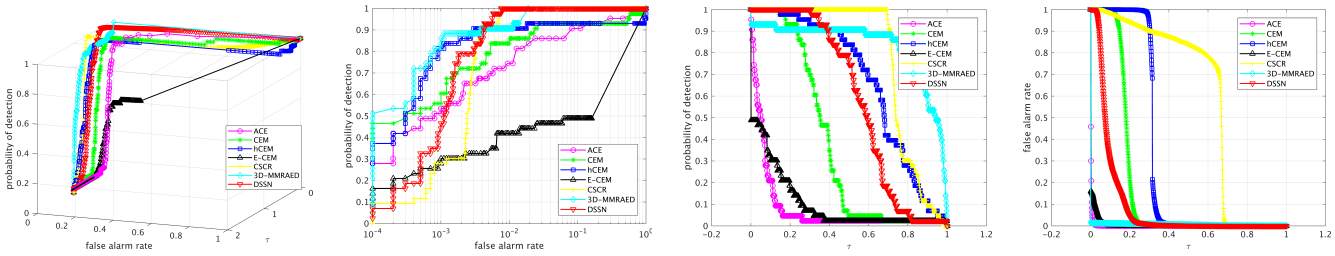

Fig. 15. 3D-ROC and the corresponding 2D-ROC curves of different algorithms for the Avon data. (a) 3D-ROC. (b) 2D-ROC of $(\mathbf{P}_f, \mathbf{P}_d)$. (c) 2D-ROC of $(\tau, \mathbf{P}_d)$. (d) 2D-ROC of $(\tau, \mathbf{P}_f)$.

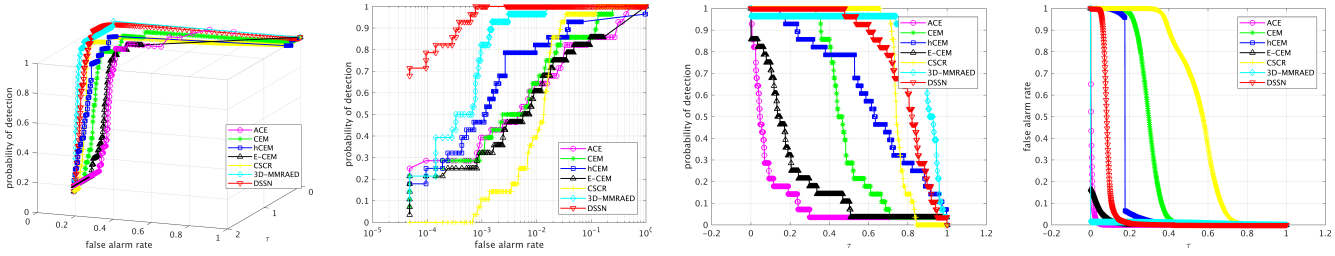

Fig. 16. 3D-ROC and the corresponding 2D-ROC curves of different algorithms for the Indian Pines data. (a) 3D-ROC. (b) 2D-ROC of $(\mathbf{P}_f, \mathbf{P}_d)$. (c) 2D-ROC of $(\tau, \mathbf{P}_d)$. (d) 2D-ROC of $(\tau, \mathbf{P}_f)$.

respectively. In addition, the inner and outer window sizes are $(5, 3)$ for the Synthetic and Avon data and $(7, 3)$ for the San Diego and Indian Pines data. For 3D-MMRAED, the optimal parameters chosen are adjusted according to [34]. For reliable assessment, 3D-MMRAED and DSSN are repeated ten times. The detection maps and running time of the two methods are the ones whose AUC is closest to the mean value, and the AUC and $\mathbf{P}_f^{\min}(\mathbf{P}_d = 1)$ values are reported in the form of mean$\pm$standard deviation.

Figs. 9–12 show the detection maps of different algorithms for the Synthetic, San Diego, Avon, and Indian Pines data sets, respectively. As seen from Figs. 9–12, there are many target pixels omitted by ACE, mainly because of interclass variation and the multivariate Gaussian distribution assumption of the background. hCEM and E-CEM aim to learn the spectral discrimination of the HSI, but there exists the problem of instability, especially E-CEM, which performs even worse than CEM by visual inspection. For CSCR, except for the detection result on the Synthetic data, all the remaining detection maps show a relatively high value on the background, making it hard to visually recognize the targets. However, the deep-learning-based 3D-MMRAED and DSSN significantly suppress the background and lead to better detection maps than others. The difference between them is that 3D-MMRAED reduces the background intensity by subsequent hierarchical suppression on the feature cube, while DSSN directly

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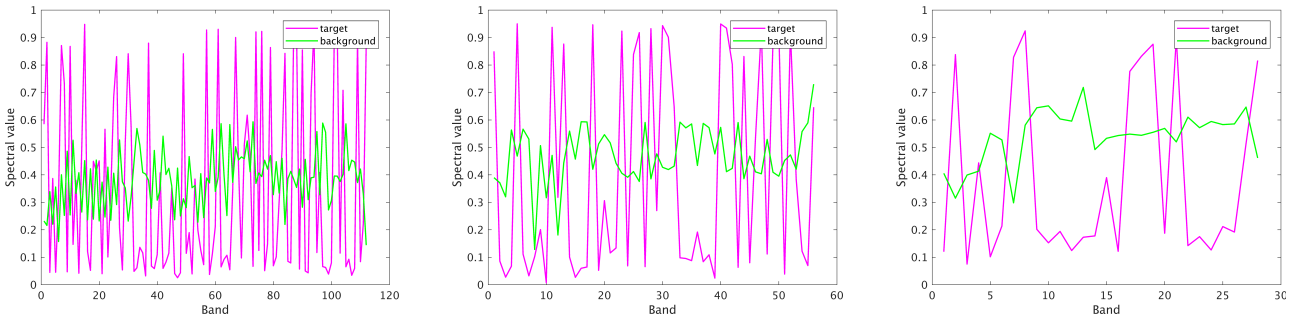

Fig. 17. Average spectral weights $\boldsymbol{\mu}_j$, $j = 1, 2, 3$ of the target and the background in three RM blocks for synthetic data. (a) $\boldsymbol{\mu}_1$. (b) $\boldsymbol{\mu}_2$. (c) $\boldsymbol{\mu}_3$.

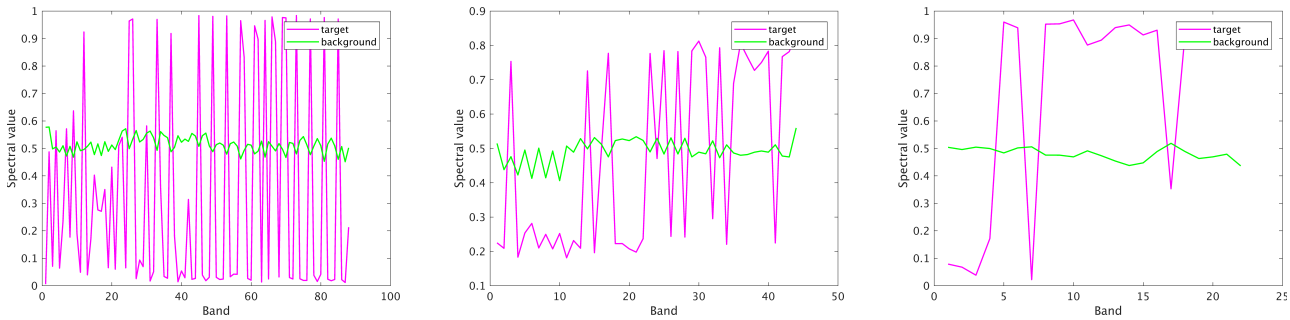

Fig. 18. Average spectral weights $\boldsymbol{\mu}_j$, $j = 1, 2, 3$ of the target and the background in the three RM blocks for San Diego data. (a) $\boldsymbol{\mu}_1$. (b) $\boldsymbol{\mu}_2$. (c) $\boldsymbol{\mu}_3$.

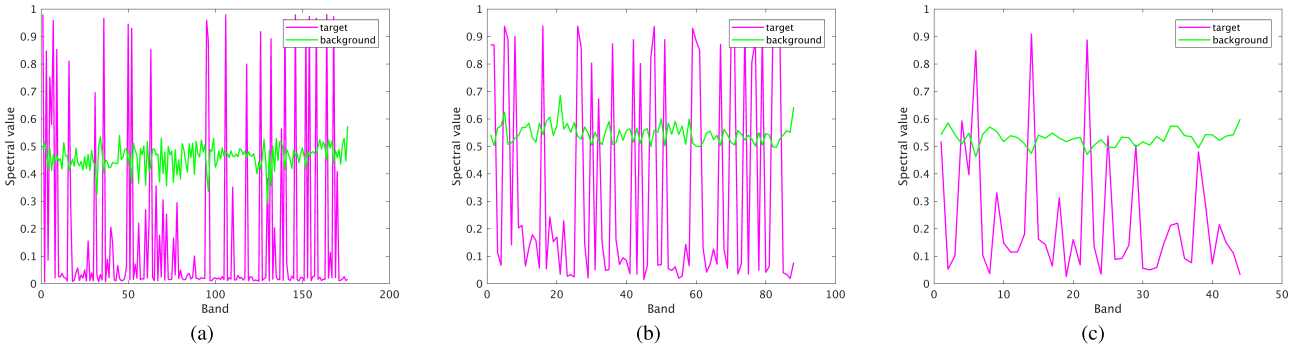

Fig. 19. Average spectral weights $\boldsymbol{\mu}_j$, $j = 1, 2, 3$ of the target and the background in three RM blocks for Avon data. (a) $\boldsymbol{\mu}_1$. (b) $\boldsymbol{\mu}_2$. (c) $\boldsymbol{\mu}_3$.

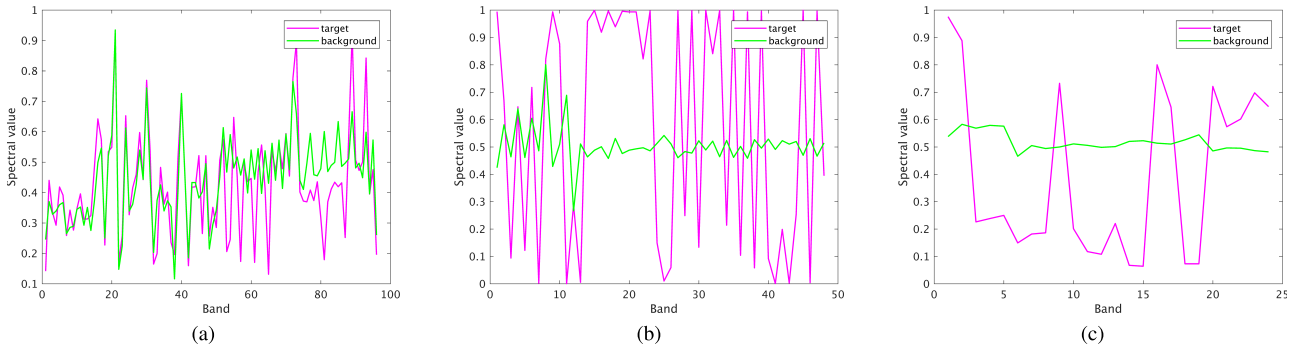

Fig. 20. Average spectral weights $\boldsymbol{\mu}_j$, $j = 1, 2, 3$ of the target and the background in three RM blocks for the Indian Pines data. (a) $\boldsymbol{\mu}_1$. (b) $\boldsymbol{\mu}_2$. (c) $\boldsymbol{\mu}_3$.

learns the intrinsic discrimination of data without any further processing. Furthermore, the Indian Pines data are used to evaluate the discrimination ability of different algorithms in these experiments. As aforesaid, the seventh class of grass-pasture-mowed is similar to the first class of alfalfa and the eighth class of hay-windrowed. From Fig. 12, we can observe that the DSSN shows less false-alarm pixels of alfalfa and hay-windrowed than CEM, hCEM, CSCR, and 3D-MMRAED, and the more correctly detected targets than the ACE and E-CEM. This demonstrated the effectiveness of the DSSN on the discriminative information preservation and useless information suppression.

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TABLE III

ACCURACY COMPARISON OF DIFFERENT METHODS FOR FOUR DATA SETS. ARROW ↑ (↓) INDICATES THAT THE LARGER (SMALLER) THE VALUE, THE BETTER THE RESULTS. BOLDFACE HIGHLIGHTS THE BEST RESULT, WHILE UNDERLINE THE SECOND

Method		ACE	CEM	hCEM	E-CEM	CSCR	3D-MMRAED	DSSN
AUC↑	Synthetic	0.9833	0.8922	0.9126	0.9564	0.9977	0.9984±0.0006	**0.9995**±0.0002
	San Diego	0.9417	0.9488	0.9714	0.8644	0.9815	0.9912±0.0007	**0.9958**±0.0006
	Avon	0.9512	0.9499	0.9305	0.7000	0.9975	0.9982±0.0005	**0.9984**±0.0004
	Indian Pines	0.9386	0.9726	0.9424	0.9066	0.9839	0.9985±0.0009	**0.9999**±0.0000
$\mathbf{P}_f^{\min}(\mathbf{P}_d = 1)\downarrow$	Synthetic	0.1491	0.6420	1.0000	0.9998	0.0095	0.0048±0.0003	**0.0010**±0.0001
	San Diego	0.9999	0.9082	0.9806	0.9999	0.4376	0.3811±0.1660	**0.0288**±0.0003
	Avon	0.9998	0.9214	0.9996	0.9998	0.0077	0.0187±0.0002	**0.0075**±0.0002
	Indian Pines	0.5232	0.2472	0.9800	1.0000	0.1233	0.0166±0.0010	**0.0008**±0.0000

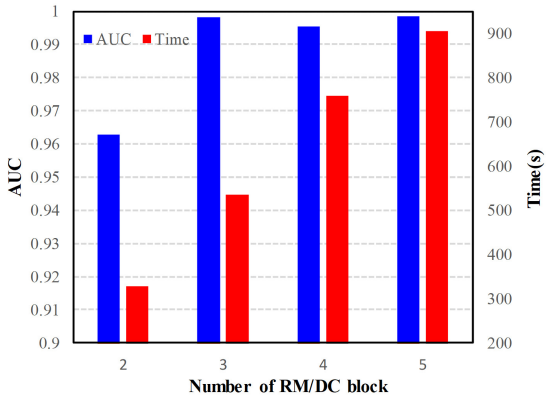

Fig. 21. Accuracy and time consumption with the number of RM/DC blocks on the Avon data set.

Evaluation solely by visual inspection is not sufficient. To further evaluate the performance of the above detection methods, 3D-ROC and three 2D-ROC curves are adopted for quantitative analysis, as shown in Figs. 13–16. It should be noted that for reliability, three 2D-ROC curves should be cooperatively used for performance evaluation, in which the 2D-ROC curve of $(\mathbf{P}_f, \mathbf{P}_d)$ is for assessing the overall detection performance, while the 2D-ROC curve of $(\tau, \mathbf{P}_d)$ and 2D-ROC curve of $(\tau, \mathbf{P}_f)$ are particularly used to evaluate the effect of target preservation and background suppression on target detection, respectively. By observing from Figs. 13(b) and 16(b), it is obvious that the DSSN showed superior detection ability to other methods for the Synthetic and Indian Pines data sets, but as for the San Diego and Avon data sets, the DSSN performed slightly worse than some methods at the low false-alarm rate. However, when $\mathbf{P}_d(\tau)$ is equal to 1, the minimal $\mathbf{P}_f(\tau)$ of the DSSN is seemingly far smaller than that of the other detectors, i.e., it was the first to achieve 100% detection probability. For example, at a P_f of 3×10^{-3}, P_d on San Diego obtained by the DSSN is about 0.4, which is slightly smaller than 0.6 and 0.8 obtained by h-CEM and 3D-MMRAED, respectively. However, hCEM and 3D-MMRAED detect all target pixels only at the false-alarm rate of around 1.0 and 0.5, respectively, while the DSSN can achieve a detection probability of 1.0 with a far lower false-alarm rate of 0.03. From this point of view, the DSSN can obtain better performance in many real applications that assign a high cost for omissive detection, such as medical diagnosis and security check. According to Figs. 13(c)–16(c), the DSSN showed superior performance over ACE, CEM, and E-CEM on four data sets. 3D-MMRAED, CSCR, and hCEM performed slightly better than the DSSN on San Diego, Avon, and Indian Pines. However, 3D-MMRAED cannot detect all target pixels until the threshold is near to zero, i.e., relatively high false-alarm rate. hCEM and CSCR obtained much worse performance than the DSSN in background suppression, as shown in Figs. 13(d)–16(d). Therefore, jointly considering target preservation and background suppression, the DSSN can achieve competitive results. Table III gives the accuracy of all competitors on four data sets. The top result is boldfaced, while the second is underlined. It can be seen that the DSSN obtained the highest AUC and the lowest $\mathbf{P}_f^{\min}(\mathbf{P}_d = 1)$ for all four data sets, indicating that the DSSN performed effectively in discriminative feature extraction and spectral variation alleviation, and demonstrated that the proposed DSSN achieved the best accuracy as a whole. Specifically, compared with CEM, the DSSN obtained an improvement of 10.7%, 4.69%, 4.84%, and 2.73% for the AUC values on the Synthetic, San Diego, Avon, and Indian Pines data sets, respectively. This indicates that the DSSN can exploit the most useful spectral–spatial information for target detection by paying more attention to the hard-detected background instead of all pixels. Moreover, if τ is appropriately selected, the DSSN could obtain $\mathbf{P}_f \approx 0$ and $\mathbf{P}_d = 1$. The main reasons that the DSSN performed better are its use of spatial–spectral RoI extraction and deep intrinsic property learning.

2) Feature Analysis: Not all the spectral features are useful for hyperspectral target detection, some of which may cause interference. Therefore, a BA mechanism is designed to select automatically the key bands with the strongest discrimination. To fully understand the BA module, we visualize and analyze the average weight of the target and background pixels in the three RM blocks for the Synthetic, San Diego, Avon, and Indian Pines data sets, as shown in Figs. 17–20. It can be observed that the average weights of the target were mostly 0 or 1, which means that the outputs of the BA module selected the useful and discarded the useless information, and were more appropriate for the target-detection task. However,

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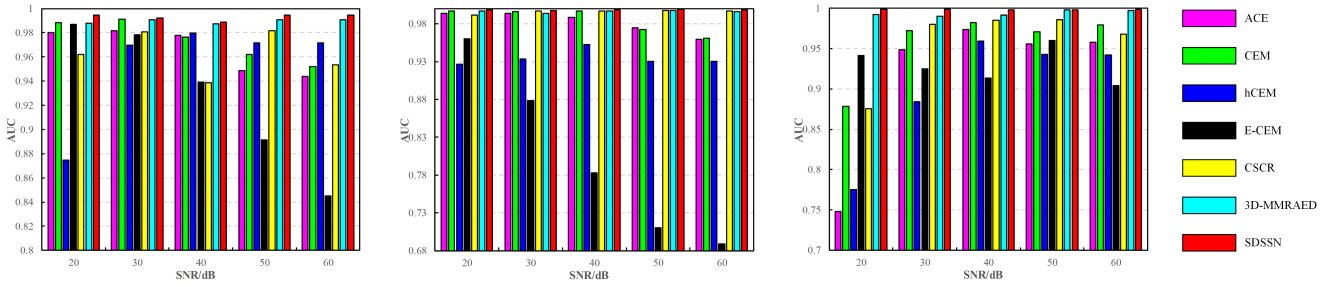

Fig. 22. Accuracy comparison on three data sets with different levels of Gaussian noise. (a) San Diego. (b) Avon. (c) Indian Pines.

for background, the average weight fluctuated around 0.5, which showed that each band was equivalent. This will further increase the distinction between the target and background pixels whose spectra are similar to the positive target pixels, i.e., helps to solve the problem of interclass similarity.

3) Effect of Number of RM/DC Block: The number of RM/DC blocks is determined by preliminary experiment and empirically set to 3 in this article, since employing less RM/DC blocks cannot achieve satisfying performance, while using more RM/DC blocks will not bring noticeable performance improvement and increase the computational cost. Fig. 21 shows the accuracy and time consumption with the number of RM/DC blocks on the Avon data. It can be seen that as the number of RM/DC blocks increases, the computational cost gradually becomes higher, and the detection performance no longer improves significantly. Thus, considering the trade-off between the detection performance and the computational complexity, we finally adopted DSSN with three RM/DC blocks.

4) Robustness to Noise: In this section, the robustness of different algorithms is evaluated on three real HSIs. The Gaussian noise with different levels of SNR (ranging from 20 to 60 dB with an interval of 10 dB) is added to obtain five noisy HSIs for each data set. Fig. 22 gives the detection accuracy of all competitors on the three data sets. From the results, we can see that the DSSN achieved the highest AUC under all circumstances and the AUC values were basically larger than 0.98, which significantly demonstrated the robustness to noise. It is also noted that 3D-MMRAED showed stability under different levels of noise, and CSCR performed well when the SNR was larger than 20 dB. In conclusion, the proposed DSSN is insensitive to noise, since the regularization term helps to avoid the network overfitting and allows small interference to the training samples.

5) Time Consumption: The detectors of E-CEM, 3D-MMRAED, and DSSN are implemented using Python 3.6 on a computer equipped with an NVIDIA GeForce GTX 1080Ti GPU, and the other competitors are carried out in MATLABR 2017a on the same computer. Table IV lists the time consumption of all testing methods on four hyperspectral data sets, in which the running time of the 3D-MMRAED and the DSSN includes the training time of the neural networks and the subsequent detection time. It can be seen that the computational complexity of the 3D-MMRAED and the DSSN is high due to the exploitation of the deep neural network, which is a typical problem in the deep-learning-based algorithms [29]. However, for the similar/same image scenes, the 3D-MMRAED and DSSN can be general enough to extract the discriminative feature without retraining, which takes only a few seconds. This characteristic plays a critical role for real-time processes. As for target detection, it is obvious that the DSSN is much faster than the other competitors, since the spectral dimension of SSFI is far less than the raw HSI. The 3D-MMRAED requires more time for detection than the DSSN, since it detects in a hierarchical way. Moreover, the DSSN achieves the highest detection accuracy among the other methods. This further verifies the superiority of the DSSN in both efficiency and accuracy.

TABLE IV
TIME CONSUMPTION OF DIFFERENT METHODS FOR FOUR DATA SETS. BOLDFACE HIGHLIGHTS THE BEST RESULT, WHILE UNDERLINE THE SECOND. 3D-MMRAED IS ABBREVIATED AS MRAE

Method		Synthetic	San Diego	Avon	Indian Pines
ACE		0.0812	0.1589	0.6024	0.2905
CEM		0.2407	0.0719	0.2483	0.1358
hCEM		0.1195	0.1132	0.1229	0.1582
E-CEM		25.8956	28.7089	48.2836	34.2237
CSCR		3.5746	3.0181	6.1478	5.6519
MRAE	train	410.8288	850.2129	531.2126	1080.5129
	detect	0.0131	0.0256	0.0229	0.0327
DSSN	train	305.4255	809.1473	558.8241	1148.9625
	detect	0.0024	0.0039	0.0033	0.0193

V. CONCLUSION

This article proposed an effective deep-learning-based method (DSSN) to extract the discriminative features for hyperspectral target detection, which requires only one positive target sample. Two novel modules named RFT and MSA are presented. Specifically, the RFT integrates the RoI map into the feature representation to extract the spatial feature of interest, and the MSA is introduced to further shed light on the ability of multiscale informative feature extraction in spectral dimension. In addition, a feature loss, composed of ℓ_1-norm and ℓ_2-norm regularizations, is presented to simultaneously preserve the target pixels and suppress the background pix-

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els. The experiments and analysis on four widely used data sets demonstrate that the proposed DSSN method is superior in performance and robustness to the other comparison algorithms.

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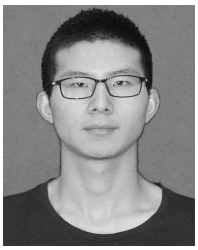
Yanzi Shi (Graduate Student Member, IEEE) received the B.S. degree from the School of Telecommunications Engineering, Xidian University, Xi'an, China, in 2017, where she is pursuing the Ph.D. degree with the State Key Laboratory of Integrated Service Networks.

Her research interests include hyperspectral target detection, feature extraction, and deep learning.

Jiaojiao Li (Member, IEEE) received the B.E. degree in computer science and technology, the M.S. degree in software engineering, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2009, 2012, and 2016, respectively.

She was an exchange Ph.D. Student with Mississippi State University, Starkville, MS, USA. She is a Post-Doctoral Researcher and a Lecturer with Xidian University. Her research interests include hyperspectral remote sensing image analysis and processing, pattern recognition, and data compression.



Yuxuan Zheng (Graduate Student Member, IEEE) received the B.E. degree from the School of Telecommunications Engineering, Xidian University, Xi'an, China, in 2017, where he is pursuing the Ph.D. degree with the State Key Laboratory of Integrated Services Networks.

His research interests include hyperspectral image fusion, hyperspectral image super-resolution, and deep learning.

Bobo Xi (Graduate Student Member, IEEE) received the B.S. degree from the School of Telecommunications Engineering, Xidian University, Xi'an, China, in 2017, where he is pursuing the Ph.D. degree with the State Key Laboratory of Integrated Service Networks.

His research interests include hyperspectral image process, machine learning, and deep learning.

Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

He joined the School of Telecommunications Engineering, Xidian University, in 1999, where he is a Professor and the Director of the Image Coding and Processing Center, State Key Laboratory of Integrated Service Networks. His research interests focus on image and video processing, hyperspectral image processing, and high-performance computing.

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